

# Esp32Tap — Depth-First Cluster Build and Test

**Guide mode:** Empty-board build · Build one cluster, prove it, sign it, then unlock the next.

No treadmill cable before clusters 1–10 pass; do not connect a treadmill cable until Cluster 11. No exact physical placement is prescribed. Follow the named parts, pins, nets, and colors.

|                           |                  |
|---------------------------|------------------|
| <b>Device serial:</b>     | <b>Operator:</b> |
| <b>Hardware revision:</b> | <b>Date:</b>     |

**Source manifest:** esp32tap-breadboard-netlist.csv; esp32tap-breadboard-clusters.html; esp32tap-breadboard-from-to.html; DigiKey invoice 129832848; superseded module checklist used only for already-proven limits and sequences.

## Source contract

**Mutually exclusive source rule:** USB and STANDALONE POWER are mutually exclusive—one source only. USB connected means STANDALONE POWER physically removed. STANDALONE POWER installed means USB physically disconnected. All sources off before any RJ45 attachment/removal, harness change, or direct-path change. COIL POWER stays open through the unloaded test.

## Safety contract

- Initial source: 8.00 V, initial 250 mA current limit; coil-open current below 50 mA. VIN must be 7.20–7.90 V and LOGIC\_3V3 must be 3.20–3.40 V.
- TREAD\_OK is low below UV, high at 8.00 V, and low above OV. UV rising is 6.25–6.55 V; OV falling is 10.30–10.90 V.
- TPS709 enabled: 4.75–5.25 V; TPS709 disabled: below 0.25 V. Loaded relay current limit no more than 500 mA.
- Coil: 90–110 mA and at least 4.50 V. BC337 VCE no more than 0.30 V. Feedback (1,0) energized and (0,1) bypass; 00 or 11 is a fault.
- Removal of USB logic power, GPIO21 command, TREAD\_OK, or VIN must restore NC bypass within 100 ms.
- Five-minute hold: TPS709 and BC337 no more than 45°C and no more than 10°C over ambient.
- Any unexpected reset is a STOP. No treadmill relay transfer or MCU TX is permitted by this guide.

**Firmware:** The current no-control diagnostic cannot pass a relay exercise gate and cannot pass a UART exercise gate. Relay/coil/TX work requires a bounded relay exerciser with exact identity. Standalone/bypass requires an observer identity and manifest proving relay and TX disabled.

Wire palette: BLACK GND · RED +8V\_RAW/FUSED\_8V · ORANGE VIN · BLUE TSR\_3V3/LOGIC\_3V3 · VIOLET +5V\_RLY/coil · YELLOW control/UART · GREEN feedback/RX · WHITE direct pass-through.

## Cluster 1 — Raw protection

**Dependencies:** Empty board and verified bench equipment  
**Inputs:** +8V\_RAW, GND

**Outputs:** VIN, FUSED\_8V  
**Source state:** USB absent; STANDALONE POWER removed;  
COIL POWER removed; bench 8.00 V, initial 250 mA limit

### Parts

**F1** RXEF075 · 750 mA hold · RXEF075HF-ND · non-polar. **D1** 1N5822-TP Schottky · 1N5822-TPMSCT-ND · cathode stripe faces VIN. **D2** P6KE12A-TP · P6KE12A-TPMSCT-ND · cathode to VIN. **CIN** 22  $\mu$ F 25 V · 732-8821-1-ND · negative stripe to GND.

**Operator part record:** F1 · RXEF075 · 750 mA hold · RXEF075HF-ND · non-polar

**Operator wiring record:** F1 · lead 1 · +8V\_RAW · RED

### Build — exact pin/net/color wiring

Wire F1 lead 1 to +8V\_RAW in RED; F1 lead 2 to FUSED\_8V in RED; D1 anode to FUSED\_8V and D1 cathode to VIN in ORANGE; D2 cathode and CIN positive to VIN in ORANGE; D2 anode and CIN negative to GND in BLACK.

### Unpowered test

| Instruction / kind / points / expected                                                                                              | Evidence |
|-------------------------------------------------------------------------------------------------------------------------------------|----------|
| Check polarity and resistance · polarity · D1 cathode/VIN, D2 cathode/VIN, CIN negative/GND, VIN–GND · stripes correct and no short |          |

### Powered test

| Instruction / input / output / evidence / limits                                                                                         | Measured |
|------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Apply protected raw power · +8V_RAW at 8.00 V · FUSED_8V and VIN · record volts/current · VIN 7.20–7.90 V; coil-open current below 50 mA |          |

**STOP gate:** Current approaches 250 mA, VIN is outside 7.20–7.90 V, polarity is wrong, or anything warms. Remove power; do not build Cluster 2.

**PASS gate:** Protection chain, polarity, VIN window, and coil-open current pass. **PASS — continue** to Cluster 2 only.

Operator:

Date:

Signed PASS:

## Cluster 2 — TSR supply

**Dependencies:** Cluster 1  
**Inputs:** VIN, GND

**Outputs:** TSR\_3V3  
**Source state:** Cluster 1 source state retained; STANDALONE POWER remains removed

### Parts

**U2** TSR 1-2433E · 3.3 V converter · 1951-TSR1-2433E-ND · pin order verified. **C2/C3** 10  $\mu$ F 50 V electrolytics · 1189-2322-ND · negative stripes to GND. **JP1** removable STANDALONE POWER · NOT ORDERED · leave removed.

**Operator part record:** U2/C2/C3/JP1 · TSR 1-2433E; two electrolytics; STANDALONE POWER · 3.3 V; 10  $\mu$ F 50 V; removable link · 1951-TSR1-2433E-ND; 1189-2322-ND; NOT ORDERED · U2 pin order verified; capacitor negatives to GND; link removed

**Operator wiring record:** U2/C2/C3/JP1 · U2 pins 1/2/3; capacitor positive/negative; removable terminals · VIN/GND/TSR\_3V3/LOGIC\_3V3 · ORANGE/BLACK/BLUE

### Build — exact pin/net/color wiring

Wire U2 pin 1 and C2 positive to VIN in ORANGE; U2 pin 2 plus both capacitor negatives to GND in BLACK; U2 pin 3 and C3 positive to TSR\_3V3 in BLUE. Fit JP1 between TSR\_3V3 and the future LOGIC\_3V3 endpoint, but leave it removed.

### Unpowered test

| Instruction / kind / points / expected                                                                                                              | Evidence |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Verify converter map and capacitor polarity · continuity · U2 pins 1/2/3 to VIN/GND/TSR_3V3 · exact mapping; TSR_3V3 not joined to future LOGIC_3V3 |          |

### Powered test

| Instruction / input / output / evidence / limits                                                                         | Measured |
|--------------------------------------------------------------------------------------------------------------------------|----------|
| Energize verified VIN · VIN · TSR_3V3 · record input/output/current · TSR_3V3 3.20–3.40 V and future LOGIC_3V3 unpowered |          |

**STOP gate:** TSR\_3V3 is outside 3.20–3.40 V, polarity is wrong, LOGIC\_3V3 is energized across the removed link, or current/temperature rises.

**PASS gate:** Verified Cluster 1 VIN produces isolated TSR\_3V3 in range. **PASS** — **continue** to Cluster 3 only.

Operator:

Date:

Signed PASS:

## Cluster 3 — DevKit and logic supply

**Dependencies:** Cluster 2; Verified USB bench source  
**Inputs:** TSR\_3V3, USB\_5V, GND

**Outputs:** LOGIC\_3V3, DevKit GND  
**Source state:** First USB-only with STANDALONE POWER removed; then USB physically disconnected before standalone link installation

### Parts

**U1** ESP32-S3-DEVKITC-1-N8R8 · 1965-ESP32-S3-DEVKITC-1-N8R8-ND · antenna clear, USB ports identified. **JP1** STANDALONE POWER · NOT ORDERED · removable and labeled. **C4** 100 nF · BC1084CT-ND · non-polar.

**Operator part record:** U1/JP1/C4 · ESP32-S3-DEVKITC-1-N8R8; STANDALONE POWER; capacitor · DevKit; removable link; 100 nF · 1965-ESP32-S3-DEVKITC-1-N8R8-ND; NOT ORDERED; BC1084CT-ND · antenna clear; USB ports identified; link removable; capacitor non-polar

**Operator wiring record:** U1/JP1/C4 · DevKit 3V3/GND; JP1 terminals; capacitor leads · LOGIC\_3V3/GND/TSR\_3V3 · BLUE/BLACK

### Build — exact pin/net/color wiring

Wire DevKit 3V3 to LOGIC\_3V3 in BLUE and DevKit GND to GND in BLACK; C4 across LOGIC\_3V3/GND. Finish JP1 between TSR\_3V3 and LOGIC\_3V3 in BLUE. Label GPIO15 jumper and GPIO21 jumper as independently removable.

### Unpowered test

| Instruction / kind / points / expected                                                                                  | Evidence |
|-------------------------------------------------------------------------------------------------------------------------|----------|
| Map source handoff · mapping · DevKit 3V3, LOGIC_3V3, TSR_3V3, JP1 · USB path mapped; JP1 removable; no source backfeed |          |

### Powered test

| Instruction / input / output / evidence / limits                                                                                              | Measured |
|-----------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Prove each exclusive source · USB_5V then TSR_3V3 · LOGIC_3V3 · record both runs · LOGIC_3V3 3.20–3.40 V; never energize USB and JP1 together |          |

**STOP gate:** Sources overlap, either source backfeeds the other, LOGIC\_3V3 is outside 3.20–3.40 V, or DevKit boot is unstable.

**PASS gate:** USB-only and standalone-only handoff both pass, with the other source physically absent. **PASS — continue** to Cluster 4 only.

Operator:

Date:

Signed PASS:

## Cluster 4 — TPS3700 voltage monitor

**Dependencies:** Cluster 1, Cluster 3  
**Inputs:** VIN, LOGIC\_3V3, GND

**Outputs:** UV\_SENSE, OV\_SENSE, TREAD\_OK, TREAD\_OK\_MCU  
**Source state:** USB logic only; STANDALONE POWER removed; bench source swept at protected VIN

### Parts

**U3** TPS3700DDCR · 296-30395-1-ND; **adapter** LCQT-SOT23-6 · A879AR-ND; **RUV** 150 k $\Omega$  · 150KXBK-ND; **ROV** 255 k $\Omega$  · 13-MFR-25F52-255K-ND; 10 k $\Omega$ /100 k $\Omega$ /4.7 k $\Omega$  · NOT ORDERED; two 1 nF · 445-173473-1-ND; 100 nF · BC1084CT-ND. Meter-map adapter pin 1.

**Operator part record:** U3/dividers/filters · TPS3700DDCR plus adapter and UV/OV networks · 150 k $\Omega$ /10 k $\Omega$ ; 255 k $\Omega$ /10 k $\Omega$ ; 1 nF; 100 nF · 296-30395-1-ND; A879AR-ND; 150KXBK-ND; 13-MFR-25F52-255K-ND; 445-173473-1-ND; BC1084CT-ND; NOT ORDERED · meter-map adapter pin 1; capacitors non-polar

**Operator wiring record:** U3/dividers/filters · U3 pins 1–6 and resistor/capacitor endpoints · VIN/GND/UV\_SENSE/OV\_SENSE/TREAD\_OK/TREAD\_OK\_MCU · ORANGE/BLACK/WHITE/YELLOW/BLUE

### Build — exact pin/net/color wiring

ORANGE: VIN to U3 pin 5, 150 k $\Omega$  top, 255 k $\Omega$  top, bypass. BLACK: U3 pin 2 and divider/filter returns. WHITE: UV\_SENSE to pin 3 from 150 k $\Omega$ /10 k $\Omega$ /1 nF; OV\_SENSE to pin 4 from 255 k $\Omega$ /10 k $\Omega$ /1 nF. YELLOW: pins 1 and 6 tied as TREAD\_OK; 10 k $\Omega$  pull-up to LOGIC\_3V3, 100 k $\Omega$  pull-down to GND, 4.7 k $\Omega$  to GPIO6 as TREAD\_OK\_MCU.

### Unpowered test

| Instruction / kind / points / expected                                                                                                                                         | Evidence |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Meter divider and adapter · resistance · VIN–UV_SENSE–GND, VIN–OV_SENSE–GND, U3 pins · 150 k $\Omega$ /10 k $\Omega$ and 255 k $\Omega$ /10 k $\Omega$ networks; pin map exact |          |

### Powered test

| Instruction / input / output / evidence / limits                                                                                                                 | Measured |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Sweep protected VIN · VIN · TREAD_OK · record low/high transitions · low below UV, high at 8.00 V, low above OV; UV rising 6.25–6.55 V; OV falling 10.30–10.90 V |          |

**STOP gate:** TREAD\_OK three-state behavior or thresholds fail; never exceed protected VIN about 11.1 V; stop if TVS draws appreciable current or warms.

**PASS gate:** Both sweep directions and GPIO6 isolated observation match limits. **PASS — continue** to Cluster 5 only.

Operator: \_\_\_\_\_ Date: \_\_\_\_\_ Signed PASS: \_\_\_\_\_

## Cluster 5 — AHC08 permission logic

**Dependencies:** Cluster 3, Cluster 4  
**Inputs:** LOGIC\_3V3, TREAD\_OK, RELAY\_CMD, TX\_ENABLE, GND

**Outputs:** RELAY\_GATE, TX\_GATE  
**Source state:** USB logic and bench VIN; remove the corresponding DevKit command jumper before manual injection

**Source state record:** USB logic and bench VIN; remove the corresponding DevKit command jumper before manual injection

### Parts

**U4** SN74AHC08N PDIP-14 · 296-4524-5-ND · notch verified, across center channel. **C5** 100 nF · BC1084CT-ND · non-polar. **R1/R2** 10 kΩ command pull-downs · NOT ORDERED.

**Operator part record:** U4/C5/R1/R2 · SN74AHC08N; capacitor; pull-downs · PDIP-14; 100 nF; two 10 kΩ · 296-4524-5-ND; BC1084CT-ND; NOT ORDERED · notch verified; capacitor non-polar

**Operator wiring record:** U4/C5/R1/R2 · U4 pins 1–14 and component endpoints · LOGIC\_3V3/GND/RELAY\_CMD/TREAD\_OK/RELAY\_GATE/TX\_ENABLE/TX\_GATE · BLUE/BLACK/YELLOW

### Build — exact pin/net/color wiring

BLUE U4 pin 14 to LOGIC\_3V3; BLACK pin 7 and unused inputs 9,10,12,13 to GND; pins 8/11 no connection. YELLOW: GPIO21/RELAY\_CMD to pin 1, TREAD\_OK pin 2, RELAY\_GATE pin 3; GPIO15/TX\_ENABLE pin 4, TREAD\_OK pin 5, TX\_GATE pin 6. Add 10 kΩ pull-downs and C5 pin 14-to-7.

### Unpowered test

| Instruction / kind / points / expected                                                                                                     | Evidence |
|--------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Check DIP orientation and control mapping · mapping · U4 pins 1–14 and named nets · exact map, unused inputs grounded, outputs not shorted |          |

### Powered test

| Instruction / input / output / evidence / limits                                                                                                                   | Measured |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Exercise both AND gates · RELAY_CMD/TX_ENABLE plus TREAD_OK · RELAY_GATE/TX_GATE · record 00,10,01,11 truth tables · output high only for command=1 and TREAD_OK=1 |          |

Remove GPIO15 jumper before manual injection of TX\_ENABLE. Remove GPIO21 jumper before manual injection of RELAY\_CMD. Never drive against a connected DevKit GPIO.

**STOP gate:** Any gate rises without both inputs, a command GPIO remains connected during manual drive, or supply/reset behavior is abnormal.

**PASS gate:** Both four-state truth tables pass and temporary drives are removed. **PASS — continue** to Cluster 6 only.

|                 |             |                    |
|-----------------|-------------|--------------------|
| Operator: _____ | Date: _____ | Signed PASS: _____ |
|-----------------|-------------|--------------------|

## Cluster 6 — TPS709 and BC337 driver

**Dependencies:** Cluster 1, Cluster 5  
**Inputs:** VIN, RELAY\_GATE, GND

**Outputs:** +5V\_RLY, RELAY\_COIL-  
**Source state:** Valid VIN/TREAD\_OK; COIL POWER remains removed for the unloaded test

### Parts

**U5** TPS70950DBVR · 296-35484-1-ND; **adapter** LCQT-SOT23-6 · A879AR-ND; **Q1** BC337-40 · 4878-BC337-40CT-ND; **RBASE** 560  $\Omega$  · MFR-25F52-560R-ND; 10 k $\Omega$  · NOT ORDERED; 1  $\mu$ F · 445-173583-1-ND; 4.7  $\mu$ F · 732-8660-1-ND; **JP2** COIL POWER · NOT ORDERED.

**Operator part record:** U5/Q1/RBASE/caps/JP2 · TPS70950DBVR; BC337-40; 560  $\Omega$ /10 k $\Omega$ ; 1  $\mu$ F/4.7  $\mu$ F; COIL POWER · 5 V regulator and unloaded driver · 296-35484-1-ND; A879AR-ND; 4878-BC337-40CT-ND; MFR-25F52-560R-ND; 445-173583-1-ND; 732-8660-1-ND; NOT ORDERED · adapter and Q1 leads mapped; electrolytic negative to GND; link removed

**Operator wiring record:** U5/Q1/RBASE/caps/JP2 · U5 pins 1–5; Q1 B/C/E; component endpoints · VIN/GND/RELAY\_GATE/+5V\_RLY/RELAY\_COIL- · ORANGE/BLACK/YELLOW/VIOLET

### Build — exact pin/net/color wiring

ORANGE VIN to U5 pin 1 and input capacitor; BLACK pin 2/cap returns/Q1 emitter; YELLOW RELAY\_GATE to U5 pin 3 and through 560  $\Omega$  to Q1 base with 10 k $\Omega$  base pull-down; U5 pin 4 no connection; VIOLET pin 5/output capacitor to +5V\_RLY, Q1 collector to RELAY\_COIL-. Fit labeled JP2 but leave removed.

### Unpowered test

| Instruction / kind / points / expected                                                                                             | Evidence |
|------------------------------------------------------------------------------------------------------------------------------------|----------|
| Verify adapter, transistor leads, and isolation · polarity · U5 pins, Q1 B/C/E, output capacitor, JP2 · exact map; COIL POWER open |          |

### Powered test

| Instruction / input / output / evidence / limits                                                                                                       | Measured |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Toggle gate unloaded · RELAY_GATE · +5V_RLY and Q1 base · record enabled/disabled/discharge · TPS709 enabled 4.75–5.25 V; TPS709 disabled below 0.25 V |          |

**STOP gate:** COIL POWER is installed, enabled or disabled limits fail, output does not discharge below 0.25 V, or Q1/U5 heats.

**PASS gate:** Unloaded regulator and base drive pass with coil isolated. **PASS — continue** to Cluster 7 only.

Operator:

Date:

Signed PASS:

## Cluster 7 — Relay coil, local contacts, and feedback

**Dependencies:** Cluster 6, Cluster 3  
**Inputs:** +5V\_RLY, RELAY\_COIL-, LOGIC\_3V3, GND

**Outputs:** LOCAL\_NC, LOCAL\_NO, FB\_NC, FB\_NO  
**Source state:** Local-only bench exercise; treadmill/RJ45 cables absent; bounded relay exerciser verified; loaded relay limit no more than 500 mA

### Parts

**K1** G5V-2 DC5 DPDT · 39-G5V-2DC5-ND · bottom-view drawing, meter-map physical pins. **D3** P6KE6.8CA bidirectional · 18-P6KE6.8CACT-ND. **JP2** COIL POWER · NOT ORDERED. **RFB** two 10 kΩ · NOT ORDERED.

**Operator part record:** K1/D3/JP2/RFB · G5V-2 DC5; P6KE6.8CA; COIL POWER; pull-ups · DPDT 5 V; bidirectional TVS; removable; two 10 kΩ · 39-G5V-2DC5-ND; 18-P6KE6.8CACT-ND; NOT ORDERED · bottom-view drawing; meter-map physical pins; TVS bidirectional

**Operator wiring record:** K1/D3/JP2/RFB · K1 pins 1/16, 4/6/8, 9/11/13 and component endpoints · +5V\_RLY/RELAY\_COIL-/LOCAL\_NC/LOCAL\_NO/GND/FB\_NC/FB\_NO · VIOLET/WHITE/BLACK/GREEN/BLUE

### Build — exact pin/net/color wiring

VIOLET JP2 from +5V\_RLY to K1 pin 1, Q1 collector/RELAY\_COIL- to pin 16, D3 across pins 1/16. WHITE local contact fixture: NC pin 4, COM pin 6, NO pin 8. BLACK feedback COM pin 11 to GND; GREEN pin 13 to GPIO4/FB\_NC and pin 9 to GPIO5/FB\_NO, each with BLUE 10 kΩ pull-up to LOGIC\_3V3.

### Unpowered test

| Instruction / kind / points / expected                                                                                                     | Evidence |
|--------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Map relay before attaching local fixture · continuity · pins 1–16, 4–6, 8–6, 13–11, 9–11 · coil about 50 Ω; NC pairs closed; NO pairs open |          |

### Powered test

| Instruction / input / output / evidence / limits                                                                                                                                                                                                                    | Measured |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Exercise bounded relay and fail sources · +5V_RLY/RELAY_COIL- · local contacts/feedback · record current, voltage, VCE, timing, temperature · coil 90–110 mA, voltage at least 4.50 V, BC337 VCE no more than 0.30 V; (1,0) energized, (0,1) bypass; 00 or 11 fault |          |

**Bound operator part record:** K1/D3/JP2/RFB · G5V-2 DC5; P6KE6.8CA; COIL POWER; pull-ups · DPDT 5 V; bidirectional TVS; removable; two 10 kΩ · 39-G5V-2DC5-ND; 18-P6KE6.8CACT-ND; NOT ORDERED · bottom-view drawing; meter-map physical pins; TVS bidirectional

**Relay exerciser identity:** \_\_\_\_\_ **Firmware identity:** \_\_\_\_\_

From energized state, removal of USB logic power must release to NC bypass; removal of GPIO21 command must release to NC bypass; removal of TREAD\_OK must release to NC bypass; removal of VIN must release to NC bypass. Each NC bypass return must occur in no more than 100 ms. Then hold five minutes: TPS709 and BC337 at or below 45°C and no more than 10°C over ambient.

**STOP gate:** Any electrical, feedback, release, reset, or five-minute thermal gate fails. Cluster 7 is local-only; it makes no end-to-end RJ45 claim.

**PASS gate:** Local contacts, feedback, current, voltage, release, and thermal evidence all pass. **PASS — continue** to Cluster 8 only.

Operator: \_\_\_\_\_ Date: \_\_\_\_\_ Signed PASS: \_\_\_\_\_



## Cluster 8 — AHC126 and UART taps

**Dependencies:** Cluster 5, Cluster 7  
**Inputs:** LOGIC\_3V3, TX\_GATE, GPIO17\_TX, CONSOLE.6, PIN3, GND

**Outputs:** TX\_DRV, GPIO18\_RX, GPIO16\_RX, LOCAL\_TX\_SELECTED  
**Source state:** Bench only, treadmill absent; bounded relay/UART exerciser identity recorded

### Parts

**U6** SN74AHC126N PDIP-14 · 296-4535-5-ND · notch verified. **C6** 100 nF · BC1084CT-ND. **R3** 100  $\Omega$  and **R4/R5** 10 k $\Omega$  · NOT ORDERED.

**Operator part record:** U6/C6/R3/R4/R5 · SN74AHC126N; capacitor; series/tap resistors · PDIP-14; 100 nF; 100  $\Omega$ ; two 10 k $\Omega$  · 296-4535-5-ND; BC1084CT-ND; NOT ORDERED · notch verified; capacitor non-polar

**Operator wiring record:** U6/C6/R3/R4/R5 · U6 pins 1–14 and resistor endpoints · LOGIC\_3V3/GND/TX\_GATE/GPIO17\_TX/TX\_DRV/CONSOLE.6/PIN3/GPIO18\_RX/GPIO16\_RX · BLUE/BLACK/YELLOW/GREEN

### Build — exact pin/net/color wiring

BLUE pin 14 to LOGIC\_3V3; BLACK pin 7 and unused pins 4,5,9,10,12,13 to GND; pins 6,8,11 no connection. YELLOW TX\_GATE pin 1, GPIO17\_TX pin 2, pin 3 through 100  $\Omega$  to TX\_DRV and K1 pin 8. GREEN CONSOLE.6 through 10 k $\Omega$  to GPIO18\_RX; PIN3 through 10 k $\Omega$  to GPIO16\_RX.

### Unpowered test

| Instruction / kind / points / expected                                                                                    | Evidence |
|---------------------------------------------------------------------------------------------------------------------------|----------|
| Check buffer and tap map · continuity · U6 pins, resistors, K1 pin 8, GPIO16/GPIO18 · exact map; disabled output isolated |          |

### Powered test

| Instruction / input / output / evidence / limits                                                                                                                                                                                              | Measured |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Exercise enable and local selection · TX_GATE/GPIO17_TX/local fixture · TX_DRV/RX taps/LOCAL_TX_SELECTED · capture logic traces · pin 3 high impedance disabled, follows pin 2 enabled; RX taps observe; relay selects TX only when energized |          |

**Relay exerciser identity:** \_\_\_\_\_ **Firmware identity:** \_\_\_\_\_

**STOP gate:** TX\_DRV drives while disabled, enabled waveform does not follow, taps disturb local lines, relay selection is wrong, or identity is absent.

**PASS gate:** High impedance, following, taps, and local relay-selected TX all pass. **PASS — continue** to Cluster 9 only.

Operator: \_\_\_\_\_

Date: \_\_\_\_\_

Signed PASS: \_\_\_\_\_

## Cluster 9 — Indicators and VBUS sensing

**Dependencies:** Cluster 3  
**Inputs:** LOGIC\_3V3, GPIO38, USB\_5V, GND

**Outputs:** POWER\_LED, STATUS\_LED, VBUS\_PRESENT\_N  
**Source state:** USB-only test then standalone-only test; never join VBUS to a local rail

### Parts

**Q2** 2N7000 · 4878-2N7000CT-ND; **D4** green LTL-4233 · 160-1130-ND; **D5** red 151051RS11000 · 732-5016-ND; 330  $\Omega$ , 1 k $\Omega$ , and 10 k $\Omega$  resistors · NOT ORDERED; 100 nF · BC1084CT-ND.

**Operator part record:** Q2/D4/D5/resistors/C7 · 2N7000; green/red LEDs; current/sense resistors; capacitor · MOSFET; 330  $\Omega$ /1 k $\Omega$ /10 k $\Omega$ ; 100 nF · 4878-2N7000CT-ND; 160-1130-ND; 732-5016-ND; BC1084CT-ND; NOT ORDERED · Q2 G/D/S mapped; LED long lead anode; flat side cathode

**Operator wiring record:** Q2/D4/D5/resistors/C7 · Q2 G/D/S, LED anode/cathode, resistor/capacitor endpoints · USB\_5V/GND/VBUS\_PRESENT\_N/LOGIC\_3V3/GPIO38/POWER\_LED/STATUS\_LED · VIOLET/BLACK/GREEN/BLUE/YELLOW

### Build — exact pin/net/color wiring

VIOLET DevKit 5V/VBUS to Q2 gate with 10 k $\Omega$  discharge to GND; BLACK Q2 source to GND; GREEN drain to GPIO7/VBUS\_PRESENT\_N with BLUE 10 k $\Omega$  pull-up to LOGIC\_3V3. YELLOW GPIO38 through 330  $\Omega$  to green LED anode; BLUE LOGIC\_3V3 through 1 k $\Omega$  to red LED anode; both cathodes BLACK to GND.

### Unpowered test

| Instruction / kind / points / expected                                                                                                            | Evidence |
|---------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Verify transistor leads and LED polarity · polarity · Q2 G/D/S, both LED anodes/cathodes · exact map; VBUS isolated from all local positive rails |          |

### Powered test

| Instruction / input / output / evidence / limits                                                                                                                                                          | Measured |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Toggle source and status · USB_5V/LOGIC_3V3/GPIO38 · VBUS_PRESENT_N/LEDs · record truth/voltage · VBUS sense active-low; USB present low, USB absent high; red power and green status indications correct |          |

**Bound operator part record:** Q2/D4/D5/resistors/C7 · 2N7000; green/red LEDs; current/sense resistors; capacitor · MOSFET; 330  $\Omega$ /1 k $\Omega$ /10 k $\Omega$ ; 100 nF · 4878-2N7000CT-ND; 160-1130-ND; 732-5016-ND; BC1084CT-ND; NOT ORDERED · Q2 G/D/S mapped; LED long lead anode; flat side cathode

VBUS sense is active-low and isolated: no connection may ever join VBUS to a local rail.

**STOP gate:** VBUS joins a local rail, active-low truth is wrong, LED polarity/current limiting is wrong, or sources overlap.

**PASS gate:** Both LEDs and isolated active-low VBUS sense pass under exclusive sources. **PASS — continue** to Cluster 10 only.

Operator:

Date:

Signed PASS:

# Cluster 10 — Whole-device standalone bench test

**Dependencies:** Cluster 1, Cluster 2, Cluster 3, Cluster 4, Cluster 5, Cluster 6, Cluster 7, Cluster 8, Cluster 9  
**Inputs:** +8V\_RAW, GND, observer firmware

**Outputs:** standalone boot, Wi-Fi observation, event log, UART idle evidence  
**Source state:** USB physically absent; STANDALONE POWER installed; observer verified; relay off; TX disabled

**Outputs record:** standalone boot · Wi-Fi observation · event log · UART idle evidence

## Parts

**ASSEMBLY** completed Clusters 1–9 · NOT ORDERED · inspect all removable links and polarities. **Firmware artifact** observer image and manifest · NOT ORDERED · exact immutable identity required.

**Operator part record:** ASSEMBLY/FW · completed Clusters 1–9; observer image and manifest · whole device; immutable diagnostic artifact · NOT ORDERED · inspect links/polarities; observer manifest proves relay and TX disabled  
**Operator wiring record:** ASSEMBLY/FW · all named inter-cluster endpoints and STANDALONE POWER terminals · +8V\_RAW/VIN/TSR\_3V3/LOGIC\_3V3/GND · RED/ORANGE/BLUE/BLACK

## Build — exact pin/net/color wiring

With all power off, inspect every named inter-cluster wire; remove USB physically; install BLUE STANDALONE POWER from TSR\_3V3 to LOGIC\_3V3; leave relay command off and TX disabled. No new uncontrolled connection is permitted.

## Unpowered test

| Instruction / kind / points / expected                                                                                                  | Evidence |
|-----------------------------------------------------------------------------------------------------------------------------------------|----------|
| Final standalone source audit · resistance · +8V_RAW/VIN/TSR_3V3/LOGIC_3V3/+5V_RLY to GND · no shorts; USB absent; source handoff exact |          |

## Powered test

| Instruction / input / output / evidence / limits                                                                                                                                                                              | Measured |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Boot observer from raw input · +8V_RAW · standalone boot/Wi-Fi/event log/UART idle · record rails, reset log, GPIO16/GPIO18 idle · VIN 7.20–7.90 V; TSR_3V3 and LOGIC_3V3 3.20–3.40 V; relay off, feedback (0,1), TX disabled |          |

**Firmware identity:** \_\_\_\_\_ **Observer identity:** \_\_\_\_\_

**Observer manifest:** \_\_\_\_\_ (path or SHA proving relay and TX outputs disabled)

**STOP gate:** Identity/manifest missing, rail out of range, reset, Wi-Fi/event-log failure, UART idle missing, relay enables, TX enables, or feedback differs from (0,1).

**PASS gate:** Identified observer boots standalone and all passive observations pass. **PASS — continue** to Cluster 11 only.

Operator: \_\_\_\_\_

Date: \_\_\_\_\_

Signed PASS: \_\_\_\_\_

## Cluster 11 — RJ45 pass-through and treadmill bypass

Cluster 11 is the end-to-end CONSOLE.6 ↔ MOTOR.6 NC bypass proof.

**Dependencies:** Cluster 7, Cluster 8, Cluster 10  
**Inputs:** CONSOLE.1–8, MOTOR.1–8, verified observer state

**Outputs:** eight mapped conductors, end-to-end CONSOLE.6 ↔ MOTOR.6 NC bypass, bypass-only treadmill evidence  
**Source state:** USB absent; standalone installed; observer verified; relay off; TX disabled; source → state → harness → direct-path sequence

### Parts

**J1/J2** CONSOLE/MOTOR RJ45 breakouts · NOT ORDERED · meter-map terminal numbers. **Harness** reviewed dual-pin fused-DMM current harness · NOT ORDERED. **K1** verified Cluster 7 relay contacts · 39-G5V-2DC5-ND.

**Operator part record:** J1/J2/HARNESS/K1 · CONSOLE/MOTOR RJ45 breakouts; dual-pin fused-DMM harness; verified relay · two 8P8C breakouts; reviewed current harness; G5V-2 · NOT ORDERED; 39-G5V-2DC5-ND · meter-map connector terminals; verify fused jack/polarity; relay bottom-view map proven

**Operator wiring record:** J1/J2/K1 · RJ45 pins 1–8; K1 pins 4/6/8 · GND/+8V\_RAW/PIN3/PIN4\_PASS/PIN5\_SAFETY/CONSOLE.6/MOTOR.6/TX\_DRV · BLACK/RED/WHITE/YELLOW

### Build — exact pin/net/color wiring

BLACK J1/J2 pins 1 and 7 to GND; RED pins 2 and 8 to +8V\_RAW; WHITE pin 3 ↔ 3, 4 ↔ 4, 5 ↔ 5; YELLOW CONSOLE.6 to K1 pin 4, MOTOR.6 to K1 pin 6, K1 pin 8 to TX\_DRV. With all sources off, map each conductor independently before joining common rails.

### Unpowered test

| Instruction / kind / points / expected                                                                                                                                                             | Evidence |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Map every connector conductor independently · mapping · CONSOLE.1–8 to MOTOR.1–8 and K1 pins 4/6/8 · each intended path below 2 Ω, unrelated paths open; end-to-end CONSOLE.6 ↔ MOTOR.6 through NC |          |

### Powered test

| Instruction / input / output / evidence / limits                                                                                                                                                                                                                                                                                                                                                | Measured |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Run bypass-only treadmill sequence · treadmill +8 V through fused harness · current/drop/receive-only UART/thermal · record source state, harness, restored direct paths, endpoints · separately measure treadmill current no more than 500 mA; supply drop no more than 50 mV; ground-return drop no more than 50 mV; 15-minute endpoints no more than 40°C and no more than 10°C over ambient |          |

**Firmware identity:** \_\_\_\_\_ **Observer identity:** \_\_\_\_\_

**Observer manifest:** \_\_\_\_\_ (path or SHA proving relay and TX outputs disabled)

### Bypass-only controlled sequence

1. All sources off before RJ45 changes. Attach cables only after Clusters 1–10 signed PASS.
2. Verify bypass-only source state: USB absent; STANDALONE POWER installed; observer verified; relay off; TX disabled.
3. With all sources off before harness changes, remove both independent direct +8V paths and install the reviewed dual-pin fused-DMM harness. Measure current separately at no more than 500 mA.
4. Remove power before removal of the harness. Verify zero current, remove harness, then restore both independent +8V paths and continuity-check each.
5. Voltage drop is measured only after direct-path restoration. Thermal test only after direct-path restoration; then run the fifteen-minute hold.

The current no-control diagnostic cannot pass the relay exercise gate and cannot pass the UART exercise gate. This is observation only. No treadmill transfer and no MCU TX. A future procedure must require qualified functional-firmware identity and matching production safety evidence; bypass PASS grants no transfer permission.

**STOP gate:** Any map, identity, current, drop, receive-only UART, bypass, thermal, or reset check fails. Power off; do not transfer the relay or transmit.

**PASS gate:** All eight paths, NC bypass, current, restored-path drop, receive-only observation, and thermal evidence pass. **PASS — continue** only to documentation closeout, never to treadmill control.

Operator: \_\_\_\_\_

Date: \_\_\_\_\_

Signed PASS: \_\_\_\_\_